

Assignment – 1

Subject: Fundamentals of Machine Learning (TCS-303)

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Part A – 1 Mark Questions

Q: Define Machine Learning.

➔ Machine Learning is the field of study that gives computers the ability to learn from data and improve performance without being explicitly programmed.

Q: Give one motivation behind using Machine Learning.

➔ ML helps in handling large, complex datasets and finding hidden patterns.

Q: Name two real-life applications of Machine Learning.

➔ Spam email detection, Self-driving cars.

Q: What is the difference between classification and regression in one sentence?

➔ Classification predicts categories (e.g., spam/not spam), while regression predicts continuous values (e.g., house price).

Q: State one example of learning associations.

➔ If a customer buys bread and butter, they are also likely to buy milk.

Q: Write one success case of Machine Learning.

➔ Google Translate – uses ML for real-time language translation.

Q: What is the origin of Machine Learning? (Give the decade/year).

➔ The term “Machine Learning” was coined by Arthur Samuel in 1959 at IBM.

Q: Define generalization in ML.

➔ Generalization is the ability of an ML model to perform well on new, unseen data.

Q: Write any one metric for evaluating a classification model.

➔ Accuracy = Correct Predictions / Total Predictions.

Q: Mention one factor to be considered before applying ML to a dataset.

➔ Data quality (clean, sufficient, and relevant data is needed).

Part B – 4 Mark Questions

Q: Explain with examples the motivation for using Machine Learning in modern society.

➡ Machine Learning is needed because traditional programming cannot handle today's massive and complex data. It can automate tasks like spam filtering in emails. It provides personalization, e.g., Amazon product recommendations. It has predictive power, e.g., forecasting stock market or weather. Thus, ML improves efficiency, saves time, and supports better decision-making.

Q: Discuss the difference between Classification and Regression with suitable examples.

➡ Classification assigns data into categories. Example: Predicting whether an email is spam or not. Regression predicts continuous values. Example: Predicting house prices based on size and location. Both are supervised learning methods, but output type is different (categorical vs numerical).

Q: What are the main steps to apply Machine Learning to data?

➡ 1. Collect and prepare data. 2. Choose an ML algorithm. 3. Train the model using training data. 4. Test and validate using unseen data. 5. Evaluate performance using metrics. 6. Deploy the model for real predictions.

Q: Explain the concept of “Abstraction and Knowledge Representation” in Machine Learning.

➡ Abstraction means simplifying real-world problems into a model that the computer can understand. Knowledge Representation is how information and patterns are stored inside the model for decision-making. Example: A decision tree stores knowledge as 'if-else' rules.

Q: Describe Apriori algorithm with an example of your choice.

➡ Apriori is used to find frequent itemsets and generate association rules. Example: In a supermarket – Support(Bread)=70%, Support(Butter)=60%. Rule: Bread → Butter, Confidence = 75%. This helps in market basket analysis (placing related items together).

Q: Discuss two success cases and one abuse/misuse of Machine Learning.

➡ Success: Healthcare diagnosis (detecting cancer using images). Success: Google Maps route optimization. Abuse: Deepfake technology can spread misinformation.

Q: What factors should be considered before applying Machine Learning algorithms to a problem?

➡ Quality and amount of data. Choice of suitable algorithm. Computation resources available. Evaluation metrics and business goal.

Q: Write short notes: (a) Generalization in ML (b) Metrics for classification evaluation

➡ (a) Generalization in ML – Ability of the model to adapt to unseen/new data instead of memorizing only training data. Good generalization avoids overfitting. (b) Metrics – Accuracy, Precision, Recall, F1-score. Example: Accuracy tells % of correct predictions.

Q: What is meant by “Learning Associations”? Give an example from daily life.

➡ It means finding patterns/relations in data. Example: If people buy laptops, they often buy laptop bags too. Used in recommendation systems and market basket analysis.

Q: Draw and explain the flow of a Machine Learning Process (from input data to prediction).

➡ Flow: Input Data → Preprocessing → Choose Algorithm → Train Model → Test Model → Evaluate → Prediction